Who Goes Cold in New York?

Heat complaints during the 2025–26 winter

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Why heat complaints?

Every year on October 1, New York City starts what it calls “heat season.” From then until May 31, landlords have to keep apartments at 68°F during the day whenever it is below 55°F outside, and at least 62°F at night no matter what the weather is. If they don’t, tenants can report it to 311, and the complaint is sent to the city’s housing department (HPD). That makes the 311 data a pretty direct record of who was cold, and when.

My first idea for this project was actually the Zillow housing data. It is clean and easy to work with, but the more I looked at it, the more it felt like I would just be redrawing the “which cities are cooling down” charts that Zillow already publishes every month. The 311 data was messier, but it let me ask questions I had to answer myself. I ended up with four questions, one for each figure:

1. When during the winter did people complain? (Figure 1)
2. How much does the weather explain it? (Figure 2)
3. Where are the complaints, and does that line up with income? (Figure 3)
4. Is it the same buildings over and over? (Figure 4)

I should say up front that the last question is not a new one. The NYC Comptroller’s office published two “Turn Up the Heat” reports (2023 and 2025) that found a small group of buildings with heat complaints year after year. I read them for background. Those reports look across several seasons and focus mainly on enforcement, so I tried to focus on just the most recent season and add the weather angle.

Data and tools

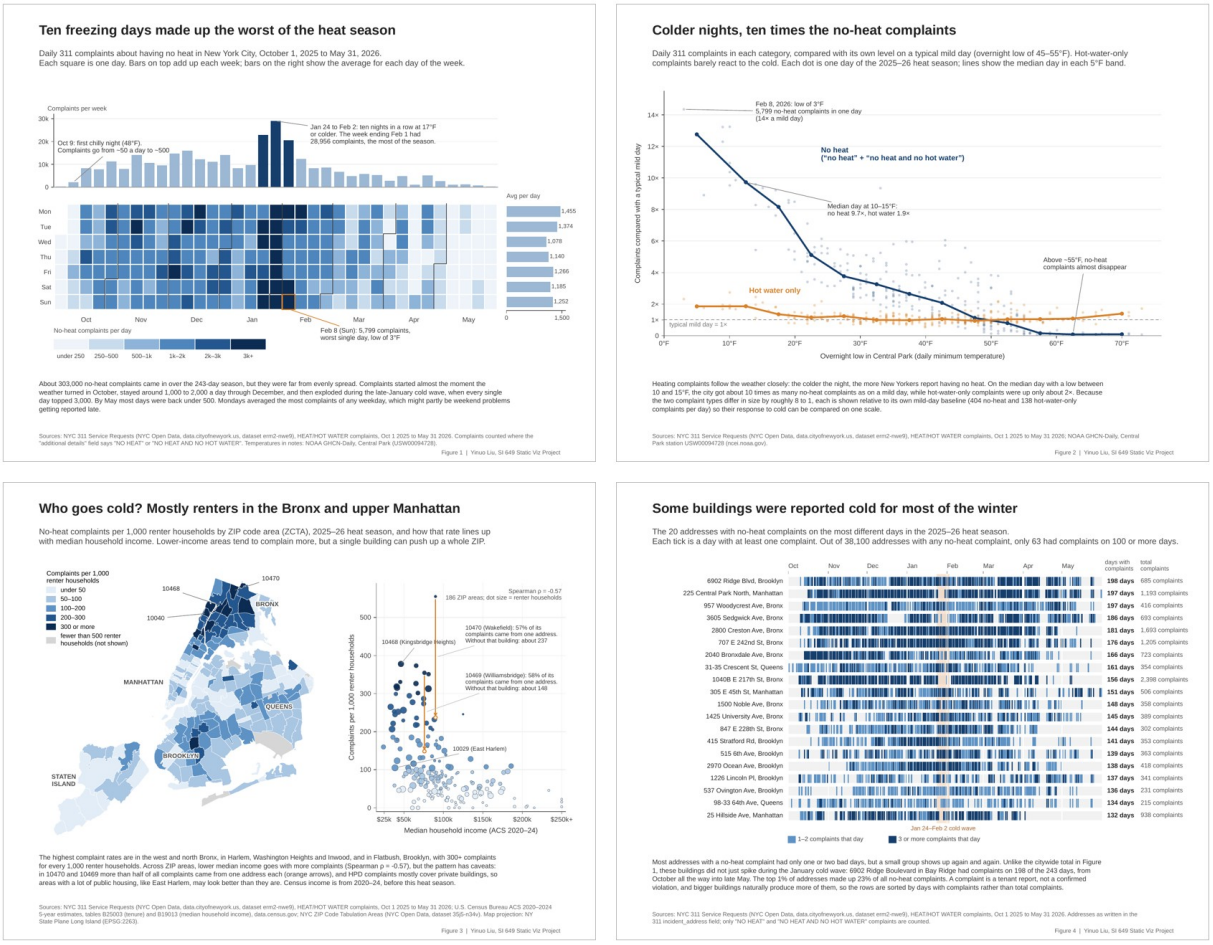
All of the complaint data comes from the NYC 311 Service Requests dataset on NYC Open Data. Instead of downloading the whole dataset (it has tens of millions of rows), I used the Socrata API to filter to complaint_type = “HEAT/HOT WATER” and created dates between October 1, 2025 and May 31, 2026. That gave me 341,824 complaints.

The most useful field turned out to be descriptor_2 (“Additional Details”), which splits the complaints into NO HEAT (211,741), NO HEAT AND NO HOT WATER (91,693) and NO HOT WATER (38,390). Every record had one of these three values, which was a nice surprise. In Figures 1, 3 and 4, a “no-heat complaint” means either of the first two categories (303,434 in total). Figure 2 compares those against the hot-water-only ones.

One thing I had to decide early was what to do about repeats. About a third of the records are another complaint from the same address on the same day (341,824 records, but only about 227,000 address-day combinations). Some of that is probably one person reporting twice, but in a big apartment building it is just as likely to be different tenants. I didn’t want to throw those away, so I kept raw counts for the citywide figures and used “days with at least one complaint” for the building-level chart, where repeats matter the most.

I did all the cleaning and plotting in Python (pandas, matplotlib and geopandas). To keep the four figures consistent, I wrote a small shared style file: one font (Liberation Sans, which is basically Arial), the same title, subtitle and caption layout on every figure, and a color rule where blue always means cold or no-heat complaints, and orange is only used for hot water and for things I want the reader to look at.

The four figures



The four final figures (each one is also submitted as its own PDF).

Figure 1 is a calendar heatmap of daily no-heat complaints. Figure 2 compares how two complaint types respond to temperature. Figure 3 maps complaint rates by ZIP code area and pairs the map with a scatter plot against median income. Figure 4 is a “barcode” chart that shows every day with a complaint for the 20 most persistent addresses. Each figure uses a different slice of the data: daily totals by date (Fig. 1), daily totals matched to NOAA weather and split by complaint type (Fig. 2), ZIP-level totals joined to Census and boundary files (Fig. 3), and address-level complaint histories (Fig. 4).

Deep dive 1: Figure 2, cold nights and complaints

Tasks

The main comparison this chart is built for is how strongly each type of complaint reacts to cold weather. A reader should be able to compare the two types on the same scale, see roughly where heating complaints start and stop (they almost disappear once the overnight low is above about 55°F), look up how big the jump is for a given temperature band (on a 10–15°F night, about 10 times a mild day), and pick out the extreme day, February 8, when the low was 3°F and the city got 5,799 complaints.

The takeaway is that heat complaints follow the weather very closely and hot-water complaints mostly don't. This doesn't prove that cold weather causes heating systems to fail, but it does show that the heat problem is very much a cold-weather problem, and that hot water behaves differently.

Data

Temperatures come from NOAA's GHCN-Daily data for the Central Park weather station (USW00094728). The file stores temperatures in tenths of a degree Celsius, so I converted them to °F. All 243 days of the season were there with no gaps. I used the daily minimum, because the overnight rule (62°F indoors) applies no matter what, and cold nights seemed like the most direct measure of how hard a heating system has to work.

There are a couple of things I'm not completely comfortable with. First, one station in Manhattan stands in for the whole city, and the outer boroughs can be a few degrees colder. Second, I'm matching complaints to the calendar day they were filed, but a lot of people probably report the morning after a cold night, so the matching is a little loose. I didn't try to shift the days around since that would just be guessing.

For the final version I also had to pick a baseline. I defined a "typical mild day" as the median of the 58 days with an overnight low between 45°F and 55°F, which is 404 no-heat complaints and 138 hot-water-only complaints. Every day is then shown as a multiple of its own baseline.

Design process

Version 1 was the obvious one: both complaint types as dots on the same axes (Figure A). It is honest about the size difference between the two groups, but it basically fails the task. The hot-water dots get squashed into a flat orange strip at the bottom, so you can't tell whether they react to temperature at all.

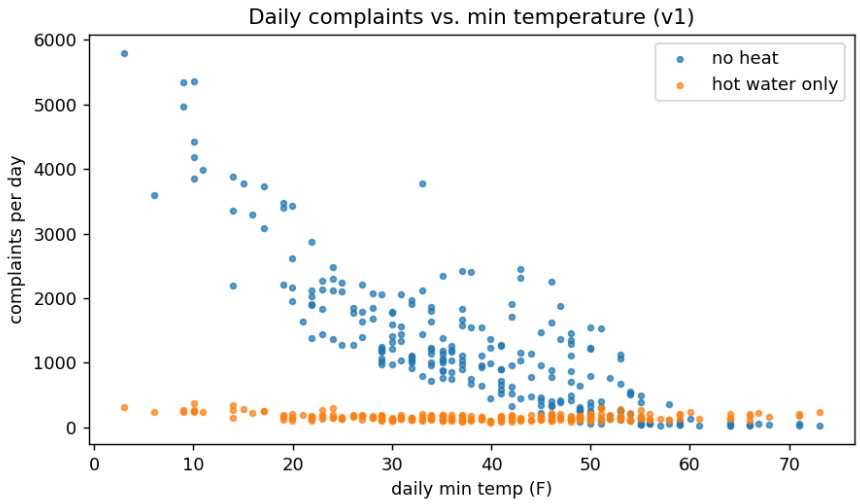


Figure A. Version 1: both complaint types on a shared y-axis.

Version 2 split them into two panels with their own y-axes, and I added a linear trend line and the Pearson correlation to each title (Figure B). This was better for seeing each shape, but it had two problems that I didn't notice right away. The linear fit for no-heat complaints goes below zero on warm days, which is impossible, so the relationship clearly isn't a straight line, and the correlation for hot water (−0.24) was hiding a U-shape where hot-water complaints go back up in late spring. The other problem is that the two panels look almost the same at a glance. You only realize that one axis goes to 6,000 and the other to 350 if you read the tick labels. In class we talked about position on a common scale being the most accurate channel, and two panels with different scales are not really a common scale.

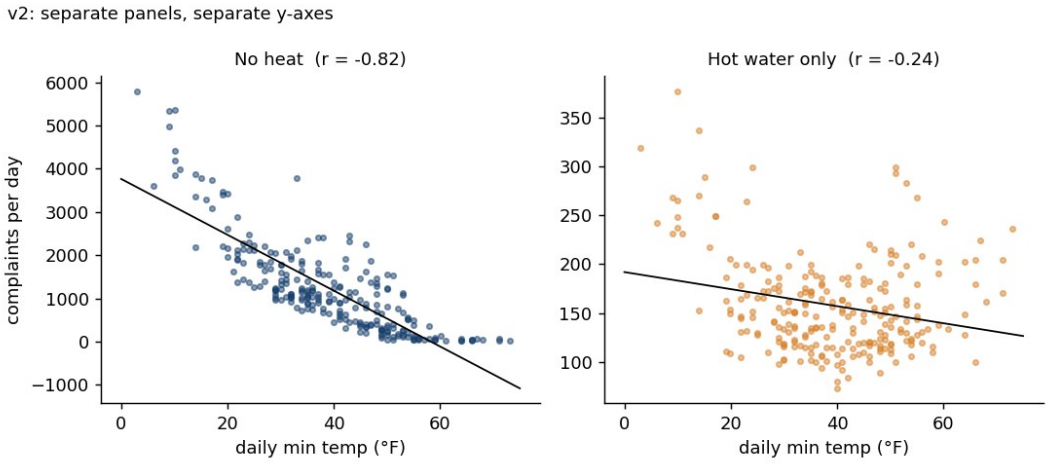
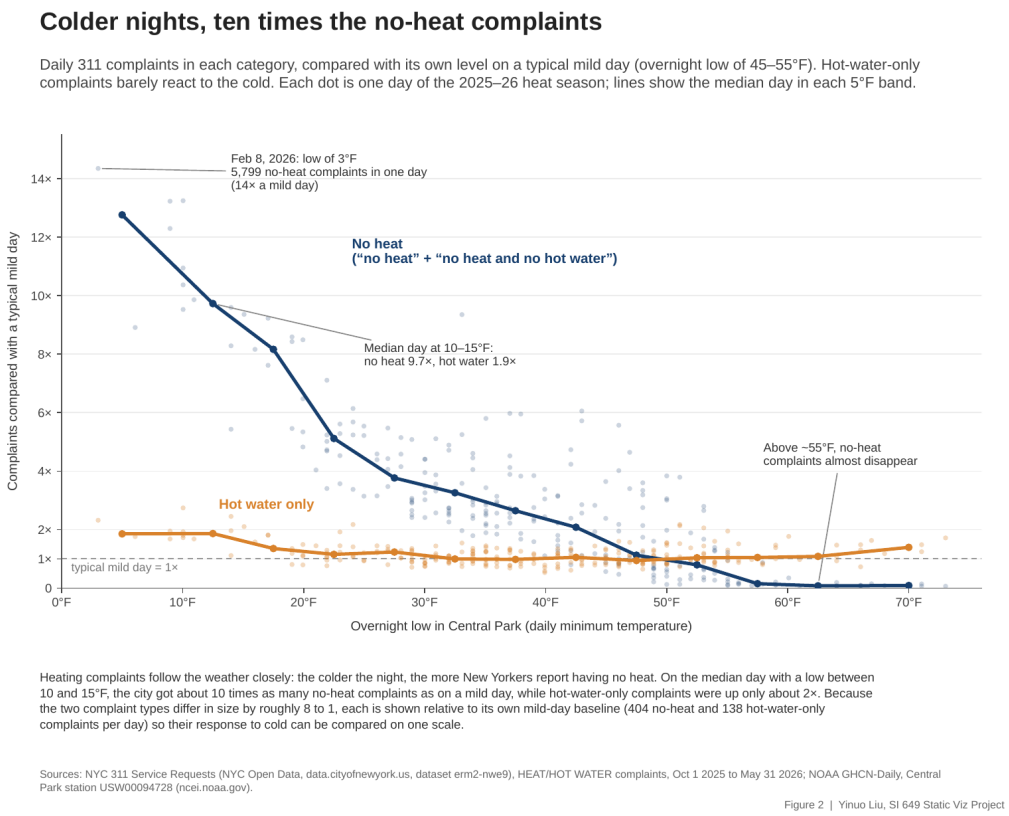


Figure B. Version 2: separate panels, separate y-axes, linear fits.

For the final version (Figure 2), I rebased each complaint type to its own mild-day level, so both can share one y-axis that reads as "times a mild day." I got this idea from the index charts that news sites use to compare stocks with very different prices. I replaced the trend line with medians in 5°F bands, which works for a curved relationship and is easy to explain. The daily dots are still there but faint, so you can see how noisy individual days are. I also took the correlation coefficient out of the main message, because "about ten times as many complaints" is easier for a general reader than "r = −0.82."

Direct labels replaced the legend, and I added three annotations (Feb 8, the 10–15°F band and the 55°F cutoff) so the key numbers can be read without going back to the axis.



Final Figure 2.

Self-evaluation

I think the final version does what I wanted: the 10× versus 2× contrast is the first thing you notice, and it is a clear improvement over both earlier versions. It follows Munzner's expressiveness and effectiveness ideas, since the main comparison is encoded with position on one shared scale. It is also fairly light on non-data ink: there are only light gridlines, no legend box and no chart border.

The weak spot is the indexing itself. "Compared with a typical mild day" takes a second to understand, and it hides the fact that there are about eight times more no-heat complaints than hot-water ones overall. I tried to cover that in the subtitle and the caption by giving the baseline numbers, but someone skimming the chart could still miss it. The coldest bands are also based on very few days (only 4 days had lows below 10°F), so the first point is less reliable than it looks. With more time, I would show the number of days in each band or add a shaded range (like the interquartile range) around the medians. It would also be interesting to use hourly temperatures instead of daily lows.

Deep dive 2: Figure 3, where the complaints are

Tasks

This figure is meant to answer "where" and "who." I wanted a reader to be able to find the neighborhoods with the highest complaint rates, compare boroughs at a glance, check whether

complaint rates have anything to do with median household income, and notice the outliers, ideally with enough context to understand why they are outliers.

The takeaway is that complaints are concentrated in the west and north Bronx, upper Manhattan and Flatbush, and that lower-income areas tend to have higher rates (Spearman $\rho = -0.57$), but also that one building can make a whole ZIP code look bad.

Data

For the rate, I divided no-heat complaints by the number of renter-occupied households in each ZIP code tabulation area (ZCTA), using the Census Bureau's ACS 2020–2024 5-year estimates (table B25003). I used renters instead of total population because heat complaints come almost entirely from renters, and dividing by population would make neighborhoods with a lot of homeowners look better than they are. Median household income comes from table B19013. I downloaded both tables from data.census.gov.

Matching places was probably the hardest part of the whole project. The 311 records have a postal ZIP code, but the Census publishes data for ZCTAs, which are area-based approximations of ZIP codes. Luckily they lined up well: out of 192 ZIP codes in my data, only three had no matching ZCTA, and those are single office-building ZIP codes with five complaints between them. For the boundaries, I first downloaded the "Modified ZCTA" file that the city's health department uses. It turned out to be based on 2010 boundaries, so 11249 in Williamsburg (which was split off from 11211 later) didn't exist in it, even though it had a few hundred complaints. I switched to NYC Open Data's regular ZCTA boundary file, which matches the 2020 ZCTAs that the ACS uses, and dropped one New Jersey ZCTA (07305, Jersey City) that was included in the file, probably because it touches the city line in the harbor.

I also had to make a few judgment calls. Thirty-four ZCTAs have fewer than 500 renter households (mostly Midtown office towers, airports and parks), and a rate there doesn't really mean anything, so I showed them in gray. Three ZCTAs list their income as "250,000+," so I plotted them at $250k and labeled that end of the axis "$250k+." The income numbers also cover 2020–2024, so they describe the neighborhoods a bit earlier than the complaints do.

Two concerns matter more for how the map should be read. HPD complaints are mostly about privately owned buildings, and public housing (NYCHA) tenants usually report problems to NYCHA instead, so places with a lot of public housing, like East Harlem, probably look better on this map than they really are. And when I looked closely at the highest-rate areas, two of them turned out to be driven by a single address: in 10470 (Wakefield) and 10469 (Williamsbridge), 57% and 58% of all no-heat complaints came from one building.

Design process

Version 1 was a map of total complaints using matplotlib's default viridis colors (Figure C). The problem was obvious once I saw it: it was mostly a map of where a lot of people live, and the brightest areas were just large, dense ZIP codes. The continuous color bar also made it hard to say what any single color meant.

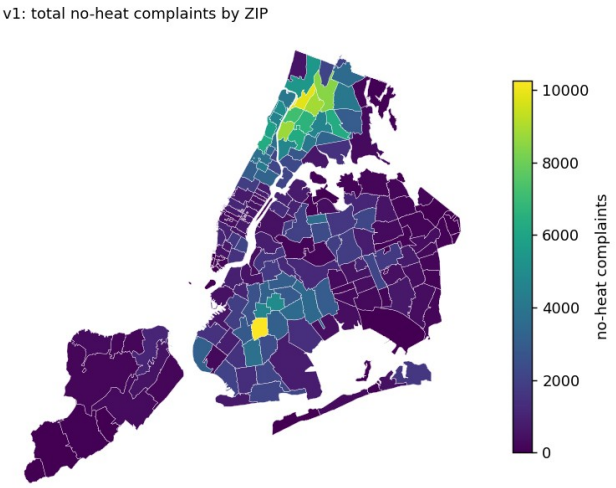


Figure C. Version 1: total complaints, default colormap.

For version 2, I switched to the rate per 1,000 renter households and tried a bivariate choropleth that combined the complaint rate and income into one 3×3 color grid (Figure D), following Joshua Stevens' guide to bivariate maps. It looked interesting, but I kept going back to the legend to remember which corner meant what, and the purple-gray colors in the middle were hard to tell apart. Splitting each variable into thirds also hid the extremes, which were the most interesting part. And there was no good way to show the single-building problem.

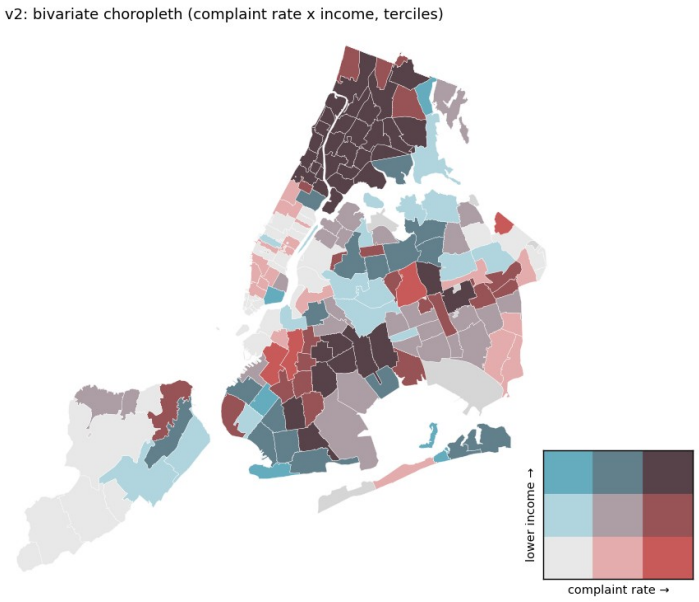
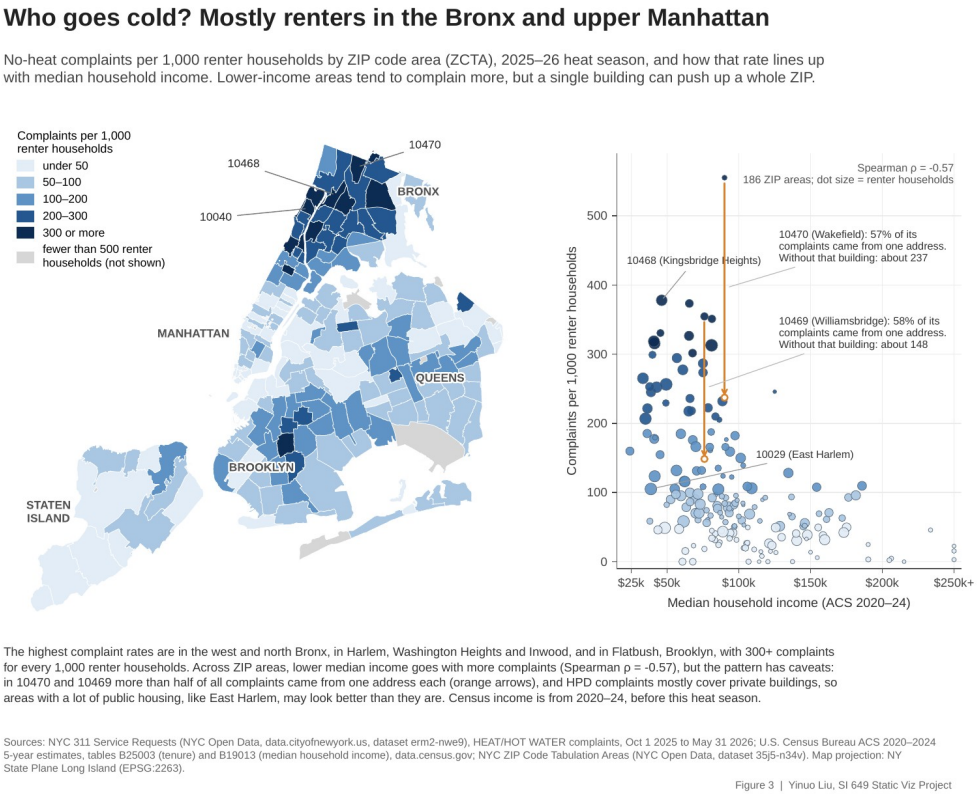


Figure D. Version 2: bivariate choropleth (complaint rate × income, in thirds).

So in the final version I split the two questions into two linked views. The map answers “where.” It uses five classes from one sequential blue ramp, from very light to navy, so darker always means more complaints. A ramp that changes mostly in lightness is easy to put in order, which was one of the main points from the color week readings. I picked the breaks by hand (50, 100, 200, 300) instead of using quantiles, so the legend numbers are round and easy to remember. Areas without enough renters are a neutral light gray so they don’t read as “low.” The map uses the New York State Plane projection (EPSG:2263) instead of plain latitude and longitude, so the city doesn’t look stretched.

The scatter plot on the right answers “who,” with income on the x-axis and the same colors as the map, so a dark dot in the scatter is a dark area on the map. Finally, for the two single-building ZIP codes, I added orange arrows showing where they would land without that one address. That way the chart is upfront about its own weak point instead of hiding it in a footnote.



Final Figure 3.

Self-evaluation

I’m happiest with the normalization and the orange arrows. Dividing by renter households turned the map from a population map into something that actually answers the question, and the arrows make a data problem part of the story. Using the same colors in both views also links them without extra explanation.

There are still real weaknesses. Choropleth maps are biased toward big areas: Staten Island and eastern Queens take up a lot of space even though few complaints come from there. Dot size in the scatter plot (renter households) is hard to compare precisely, because people tend to underestimate differences in area (Stevens’ power law), so I treat size as background information only. ZIP code areas are also large

and hide differences between blocks. The 500-renter cutoff is a judgment call, and the public housing issue is only explained in the caption, not shown on the map. With more time, I would try a building-level dot map and add NYCHA developments as an overlay, so readers could see where the data is missing.

Notes on Figures 1 and 4

Figure 1 (calendar heatmap). The idea came from GitHub's contribution graph. Daily counts are very skewed (from 26 to 5,799), so a continuous color scale made most days look the same, and I used six hand-picked steps instead. Since color isn't great for reading exact amounts, I added two sets of marginal bars: weekly totals on top and the average for each weekday on the right. The thin dark lines mark month boundaries. The chart shows the late-January cold wave very clearly, plus a small side finding that Mondays are the busiest weekday.

Figure 4 (barcode chart). My first try was a ranked dot plot of total complaints per address, but that mostly ranked buildings by how big they are. The number of days with at least one complaint is a better measure of how long a problem lasted, and drawing each of those days as a tick (a bit like Tufte's sparklines) also shows when it happened. The light orange band marks the same cold wave as in Figure 1, and the contrast is the most interesting part: the citywide total spikes in late January, but these buildings had complaints from October all the way into May.

Looking back

Looking at the four figures together, I think the style is consistent, and the story moves in a clear order: when, why, where, and which buildings. If I kept working on this, the main thing I would add is the HPD violations data, because right now everything is based on complaints, and a complaint is not proof that a building broke the law. I would also like to repeat Figure 4 over several winters, which would make it directly comparable to the Comptroller's multi-season analysis. One color decision I'm still not sure about is that orange means "hot water" in Figure 2 but "look here" in the other three figures. The two meanings never clash within one figure, but a stricter style would probably give each color only one job.

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